

TIØ4145 Corporate Finance

Exercise 11: Guest Lecture

Deadline: Monday November 17th, 2025, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday November 13th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on November 10th.

Task 1: Risk Neutral Valuation

We look at a simplified insurance product:

- 10 years time horizon. Only cash flows at $T=10$.
- Policyholder gets guaranteed return (2%, annual discounting). Guaranteed payout: 121.90.
- Deficits at $T=10$ have to be covered by insurance company.
- Initial asset value=100. Assets invested in equity index.
- At $T=10$ the policyholder gets 80% of excess profits and the insurance company 20%.
- 11 risk neutral economic scenarios have been generated.

Question:

- What is the market value of this product?

Task 2: Martingale testing

In the Excel sheet you will find:

- 1.000 scenarios of the value of an equity investment for years 0, 1, 2, ..., 10.
- 1.000 scenarios of the value of the Bank Account/discount factor $e^{\int_0^T r_t ds}$ for years 0, 1, 2, ..., 10.

- The risk free discount curve at $t=0$ for maturities 1, 2, ..., 10.

Question:

- Do a martingale test for the equity investment at times 1, 2, ..., 10 using the provided scenarios. (See example 1 on Slide 29 in the Power Point presentation from the guest lecture.)
- What do you conclude?